Skill Activity 1 — Design a relational database for a chosen domain

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CREATE DATABASE finance_db;
USE finance_db;

CREATE TABLE customers (
  customer_id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE,
  phone VARCHAR(15) UNIQUE,
  date_of_birth DATE NOT NULL
);
CREATE TABLE account_types (
  account_type_id INT AUTO_INCREMENT PRIMARY KEY,
  type_name VARCHAR(50) NOT NULL UNIQUE,
  interest_rate DECIMAL(5,2) NOT NULL DEFAULT 0.00,

  CHECK (interest_rate >= 0)
);
CREATE TABLE accounts (
  account_id INT AUTO_INCREMENT PRIMARY KEY,
  customer_id INT NOT NULL,
  account_type_id INT NOT NULL,
  account_number VARCHAR(20) NOT NULL UNIQUE,
  balance DECIMAL(12,2) NOT NULL DEFAULT 0.00,
  opened_date DATE NOT NULL,

  CONSTRAINT fk_account_customer
    FOREIGN KEY (customer_id)
    REFERENCES customers(customer_id),

  CONSTRAINT fk_account_type
    FOREIGN KEY (account_type_id)
    REFERENCES account_types(account_type_id),

  CHECK (balance >= 0)
);
```

```
CREATE TABLE categories (
  category_id INT AUTO_INCREMENT PRIMARY KEY,
  category_name VARCHAR(50) NOT NULL UNIQUE
);
CREATE TABLE transactions (
  transaction_id INT AUTO_INCREMENT PRIMARY KEY,
  account_id INT NOT NULL,
  category_id INT NOT NULL,
  transaction_type VARCHAR(10) NOT NULL,
  amount DECIMAL(12,2) NOT NULL,
  transaction_date DATE NOT NULL,
  description VARCHAR(255),

  CONSTRAINT fk_transaction_account
    FOREIGN KEY (account_id)
    REFERENCES accounts(account_id),

  CONSTRAINT fk_transaction_category
    FOREIGN KEY (category_id)
    REFERENCES categories(category_id),

  CHECK (transaction_type IN ('income', 'expense')),
  CHECK (amount > 0)
);


INSERT INTO customers
(name, email, phone, date_of_birth)
VALUES
('Aarav Sharma', 'aarav@gmail.com', '9876543210', '2001-05-12'),
('Priya Patel', 'priya@gmail.com', '9876543211', '2000-08-20'),
('Rahul Mehta', 'rahul@gmail.com', '9876543212', '1999-11-15'),
('Sneha Joshi', 'sneha@gmail.com', '9876543213', '2002-03-25'),
('Vikram Singh', 'vikram@gmail.com', '9876543214', '1998-07-10');

INSERT INTO account_types
(type_name, interest_rate)
VALUES
('Savings', 4.00),
('Current', 0.00),
('Fixed Deposit', 6.50);
```

```
INSERT INTO accounts
(customer_id, account_type_id, account_number, balance, opened_date)
VALUES
(1, 1, 'ACC10001', 50000.00, '2024-01-10'),
(2, 1, 'ACC10002', 75000.00, '2024-02-15'),
(3, 2, 'ACC10003', 120000.00, '2024-03-20'),
(4, 1, 'ACC10004', 45000.00, '2024-04-05'),
(5, 3, 'ACC10005', 200000.00, '2024-05-12');

INSERT INTO categories
(category_name)
VALUES
('Salary'),
('Food'),
('Rent'),
('Shopping'),
('Utilities'),
('Investment');

INSERT INTO transactions
(account_id, category_id, transaction_type, amount, transaction_date, description)
VALUES
(1, 1, 'income', 60000.00, '2024-06-01', 'Monthly salary'),
(1, 2, 'expense', 2500.00, '2024-06-03', 'Groceries and food'),
(1, 3, 'expense', 15000.00, '2024-06-05', 'Monthly rent'),
(2, 1, 'income', 55000.00, '2024-06-01', 'Monthly salary'),
(2, 4, 'expense', 5000.00, '2024-06-10', 'Clothing and shopping'),
(2, 5, 'expense', 3000.00, '2024-06-12', 'Electricity bill'),
(3, 1, 'income', 80000.00, '2024-06-01', 'Monthly salary'),
(3, 6, 'expense', 10000.00, '2024-06-15', 'Investment contribution'),
(4, 1, 'income', 50000.00, '2024-06-01', 'Monthly salary'),
(4, 2, 'expense', 3000.00, '2024-06-18', 'Food expenses'),
(5, 6, 'income', 15000.00, '2024-06-20', 'Investment return');

SELECT * FROM customers;

# View all accounts
SELECT
  a.account_id,
  c.name AS customer_name,
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```
  at.type_name AS account_type,
  a.account_number,
  a.balance,
  a.opened_date
FROM accounts a
JOIN customers c
  ON a.customer_id = c.customer_id
JOIN account_types at
  ON a.account_type_id = at.account_type_id;

# View transactions with account and category
SELECT
  t.transaction_id,
  c.name AS customer_name,
  a.account_number,
  cat.category_name,
  t.transaction_type,
  t.amount,
  t.transaction_date,
  t.description
FROM transactions t
JOIN accounts a
  ON t.account_id = a.account_id
JOIN customers c
  ON a.customer_id = c.customer_id
JOIN categories cat
  ON t.category_id = cat.category_id;

# verifying integrity
# constraint violation 1
INSERT INTO transactions
(account_id, category_id, transaction_type, amount, transaction_date, description)
VALUES
(1, 1, 'income', -5000.00, '2024-07-01', 'Invalid transaction');

# error output:
Error Code: 3819. Check constraint 'transactions_chk_2' is violated.

# constraint violation 2
INSERT INTO customers
(name, email, phone, date_of_birth)
```

```
VALUES
('Test User', 'aarav@gmail.com', '9999999999', '2000-01-01');

# error output:
Error Code: 1062. Duplicate entry 'aarav@gmail.com' for key 'customers.email'
```